

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Pharm. Examination

University Backlog Theory Examination March- 2018

PH114 & PH123 Pharmaceutical Chemistry-I [New Syllabus]

Date: 27/03/2018, Tuesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

Q. 1 Attempt all questions. Each question is of one mark.

20

1. Which of the following is not used in limit test of chloride?
[A] Dilute HCl
[B] Dilute HNO₃
[C] AgNO₃
[D] None of the above
2. Which is the right formula for a ring detachment method for measurement of surface tension?
[A] $\gamma = \frac{F}{4\pi}$
[B] $\gamma = \frac{F}{2\pi}$
[C] $\gamma = F \cdot 2\pi r$
[D] $\gamma = F \cdot 4\pi r$
3. Sodium chloride is prepared from_____.
[A] Sodium carbonate
[B] Sodium Phosphate
[C] Magnesium carbonate
[D] None of the above
4. Which of the following is not used as electrolyte replenisher?
[A] Sodium chloride
[B] Talc
[C] Potassium chloride
[D] Magnesium Sulphate

5. The meaning of Cool as per IP is _____ °C temperature range.
[A] 2 to 8
[B] 8 to 25
[C] 25 to 34
[D] More than 45
6. Hypokalemia means:
[A] Higher serum potassium level than that of normal
[B] Same serum potassium level as that of normal
[C] Lower serum potassium level than that of normal
[D] None
7. In _____ limit test, we are comparing a colour instead of turbidity.
[A] Chloride
[B] Lead
[C] Iron
[D] Sulphate
8. Calcium Gluconate is assayed by which method?
[A] Complexometric titration
[B] Nitrite titration
[C] Mohr's method
[D] None of the above
9. Intracellular fluid constituent _____.
[A] 30-50% of body weight
[B] 45-50% of body weight
[C] 12-15% of body weight
[D] 4-5 % of body weight
10. Radioactive Sodium is used in diagnosis of _____.
[A] Carbon dating
[B] Blood circulation
[C] Thyroid function
[D] Leukemia
11. One of the following is the drug of choice in cyanide poisoning:
[A] Sodium nitrate
[B] Sodium Thiosulfate
[C] Sodium nitrate and Sodium Thiosulfate Both
[D] None of above
12. The substances which initiate a photochemical reaction but itself does not undergo any chemical change is called _____.
[A] catalysis
[B] fluorescent
[C] sensitizer
[D] none of these

13. The role of citric acid in limit test of iron is:
[A] Prevent supersaturation
[B] Complex formation with Fe- ion metal
[C] Complex Formation with other than Fe-Ion metal
[D] Provide a Acidity
14. In photochemical reactions, the absorption of light takes place in:
[A] primary processes only
[B] secondary processes only
[C] either primary or secondary process
[D] both primary and secondary processes
15. The boiling point of a liquid is that temperature at which _____.
[A] the vapour pressure of the liquid is equal to the atmospheric pressure
[B] the vapour pressure of the liquid is less than the atmospheric pressure
[C] the vapour pressure of the liquid is greater than the atmospheric pressure
[D] the vapour pressure of the liquid is equal to the square root of the atmospheric pressure
16. Graham's law refers to:
[A] Boiling point of water
[B] Volume changes of gases
[C] Gas compression
[D] Gaseous diffusion
17. Impurities in pharmaceutical preparation may be due to following sources:
[A] Raw material
[B] Manufacturing process
[C] Chemical instability
[D] All of the above
18. Nature of Boric Acid is:
[A] Weak Acid
[B] Strong Acid
[C] Neutral
[D] Weak Base
19. Mathematically, the phase rule can be expressed by:
[A] $F + P = C + 2$
[B] $F = C - P + 2$
[C] $P = C - F + 2$
[D] All of above
20. The force per unit length over the surface of liquid is known as:
[A] Viscosity
[B] Surface tension
[C] Osmotic pressure
[D] Gravitational pull

SECTION – II

- Q 2 Attempt any TWO of the following:**
- A** Enlist sources of impurities in pharmaceutical substance. Discuss any two. **05**
- B** Write a note on ORS. **05**
- C** Write the principle, reactions and procedure involved in the limit test for chloride **05**
- Q 3 Attempt any TWO of the following:**
- A** Give the method of preparation and uses of Ferrous Sulphate. **05**
- B** Write a note on physiological acid base balance. **05**
- C** Explain importance of Sodium as an electrolyte. **05**

SECTION – III

- Q 4 Attempt any FOUR of the following:**
- A** Define the following terms (**Any Five**): **05**
(1) Antacid, (2) Viscosity, (3) Expectorant, (4) Refractive Index, (5) Cathartics, (6) Optical Rotation and (7) Interstitial Fluid
- B** Explain applications of radioactive substances in pharmaceuticals. **05**
- C** Write a note on Joule-Thomson effect. **05**
- D** Describe physiological importance of Magnesium in detail. **05**
- E** Describe preparation, assay and storage condition for Oxygen as important gas in pharmacy. **05**
- F** What is Surface Tension? Describe any TWO methods for determination of Surface Tension. **05**
- Q 5 Attempt any FOUR of the following:**
- A** Write a short note on Dentifrices. **05**
- B** Explain Jablonski diagram of photochemistry. **05**
- C** Differentiate Ideal Solution and Real solution in detail. **05**
- D** State mechanism of antimicrobial agents. **05**
- E** Short note on Debye Huckel theory. **05**
- F** What is Pharmacopoeia? Mention the contents of official monograph as per Pharmacopoeia. **05**
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